

2/20/2026

PRÁCTICA 09

Get-Post

MAESTRA: DIANA XÓCHITL RUANO VARGAS

MATERIA: APLICACIONES WEB

DIEGO EFRAÍN HERNÁNDEZ HERNÁNDEZ

TIID_04_01

ENERO - ABRIL 2026

MONGO

MongoDB Compass - upp_db/escuelaDB.alumnos

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

upp_db

- admin
- cartelera_cultural
 - eventos
- config
- escuelaDB
 - alumnos
- local

cartelera_cultural

alumnos

+

upp_db > escuelaDB > alumnos

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Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

UPDATE

DELETE

EXPORT DATA

EXPORT CODE

25

1 - 4 of 4

_id: ObjectId('6998889e78219561c6d2cd59')

nombre: "Efra GG"

edad: 21

carrera: "TIID"

_id: ObjectId('699888b978219561c6d2cd5b')

nombre: "Pacap"

edad: 19

carrera: "TIID"

_id: ObjectId('699888cd78219561c6d2cd5d')

nombre: "JLuis Smash"

edad: 22

carrera: "TIID"

VISUAL BACKEND

```
python app.py insertar_alumno
1 from flask import Flask, request, jsonify
2 from flask_cors import CORS
3 from pymongo import MongoClient
4
5 app = Flask(__name__)
6 CORS(app)
7
8 client = MongoClient("mongodb://localhost:27017/")
9 db = client["escuelaDB"]
10 coleccion = db["alumnos"]
11
12 @app.route('/')
13 def inicio():
14     return "API UPP funcionando"
15
16 @app.route('/alumnos', methods=['GET'])
17 def obtener_alumnos():
18     alumnos = list(coleccion.find({}, {"_id": 0}))
19     return jsonify(alumnos)
20
21 @app.route("/alumnos", methods=['POST'])
22 def insertar_alumno():
23     datos = request.json
24     coleccion.insert_one(datos)
25     return jsonify({"mensaje": "Alumno insertado correctamente"})
26
27 if __name__ == '__main__':
28     app.run(debug=True)
```

```
python app.py
PS C:\Users\shado\OneDrive\Documentos\GitHub\Practicas_web_Diego_Efrain_Hernandez_Hernandez\Practica 09\backend> python -m pip install flask
Requirement already satisfied: flask in C:\Users\shado\AppData\Roaming\Python\Python314\site-packages (3.1.2)
Requirement already satisfied: pymongo in C:\Users\shado\AppData\Roaming\Python\Python314\site-packages (4.16.0)
Requirement already satisfied: flask-cors in C:\Users\shado\AppData\Roaming\Python\Python314\site-packages (6.0.2)
Requirement already satisfied: blinker>=1.9.0 in C:\Users\shado\AppData\Roaming\Python\Python314\site-packages (from flask) (1.9.0)
Requirement already satisfied: click>=8.1.3 in C:\Users\shado\AppData\Roaming\Python\Python314\site-packages (from flask) (8.3.1)
Requirement already satisfied: itsdangerous>=2.2.0 in C:\Users\shado\AppData\Roaming\Python\Python314\site-packages (from flask) (2.2.0)
Requirement already satisfied: jinja2>=3.1.2 in C:\Users\shado\AppData\Roaming\Python\Python314\site-packages (from flask) (3.1.6)
Requirement already satisfied: markupsafe>=2.1.1 in C:\Users\shado\AppData\Roaming\Python\Python314\site-packages (from flask) (3.0.3)
Requirement already satisfied: werkzeug>=3.1.0 in C:\Users\shado\AppData\Roaming\Python\Python314\site-packages (from flask) (3.1.5)
Requirement already satisfied: dnspython<3.0.0,>=2.6.1 in C:\Users\shado\AppData\Roaming\Python\Python314\site-packages (from pymongo) (2.8.0)
Requirement already satisfied: colorama in C:\Users\shado\AppData\Roaming\Python\Python314\site-packages (from click>=8.1.3->flask) (0.4.6)
PS C:\Users\shado\OneDrive\Documentos\GitHub\Practicas_web_Diego_Efrain_Hernandez_Hernandez\Practica 09\backend> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 554-382-332
127.0.0.1 - - [20/Feb/2026 10:23:01] "GET / HTTP/1.1" 200 -
* Detected change in 'C:\Users\shado\OneDrive\Documentos\GitHub\Practicas_web_Diego_Efrain_Hernandez_Hernandez\Practica 09\backend\app.py', reloading
* Restarting with stat
* Debugger is active!
* Debugger PIN: 554-382-332
127.0.0.1 - - [20/Feb/2026 10:50:35] "GET / HTTP/1.1" 200 -
PS C:\Users\shado\OneDrive\Documentos\GitHub\Practicas_web_Diego_Efrain_Hernandez_Hernandez\Practica 09\backend> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 554-382-332
127.0.0.1 - - [20/Feb/2026 11:00:29] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [20/Feb/2026 11:03:04] "GET /alumnos HTTP/1.1" 200 -
127.0.0.1 - - [20/Feb/2026 11:03:41] "OPTIONS /alumnos HTTP/1.1" 200 -
127.0.0.1 - - [20/Feb/2026 11:03:41] "POST /alumnos HTTP/1.1" 200 -
```

VISUAL BACKEND PRUEBA



VISUAL FRONTEND

```
PS C:\Users\shado\OneDrive\Documentos\GitHub\Practicas_web_Diego_Efrain_Hernandez_Hernandez\Practica 09> ng new frontend
✓ Which stylesheet system would you like to use? CSS [ https://developer.mozilla.org/docs/Web/CSS ]
✓ Do you want to enable Server-Side Rendering (SSR) and Static Site Generation (SSG/Prerendering)? Yes
✓ Which AI tools do you want to configure with Angular best practices? https://angular.dev/ai/develop-with-ai None
CREATE frontend/angular.json (2142 bytes)
CREATE frontend/package.json (1184 bytes)
CREATE frontend/README.md (1520 bytes)
CREATE frontend/tsconfig.json (990 bytes)
CREATE frontend/.editorconfig (331 bytes)
CREATE frontend/.gitignore (666 bytes)
CREATE frontend/tsconfig.app.json (464 bytes)
CREATE frontend/tsconfig.spec.json (456 bytes)
CREATE frontend/.vscode/extensions.json (134 bytes)
CREATE frontend/.vscode/launch.json (490 bytes)
CREATE frontend/.vscode/mcp.json (191 bytes)
CREATE frontend/.vscode/tasks.json (1020 bytes)
CREATE frontend/src/main.ts (228 bytes)
CREATE frontend/src/index.html (307 bytes)
CREATE frontend/src/styles.css (81 bytes)
CREATE frontend/src/main.server.ts (300 bytes)
CREATE frontend/src/server.ts (1677 bytes)
CREATE frontend/src/app/app.spec.ts (698 bytes)
CREATE frontend/src/app/app.ts (302 bytes)
CREATE frontend/src/app/app.css (0 bytes)
CREATE frontend/src/app/app.html (20446 bytes)
CREATE frontend/src/app/app.config.ts (453 bytes)
CREATE frontend/src/app/app.routes.ts (80 bytes)
CREATE frontend/src/app/app.config.server.ts (438 bytes)
CREATE frontend/src/app/app.routes.server.ts (174 bytes)
CREATE frontend/public/favicon.ico (15086 bytes)
✓ Packages installed successfully.
  Directory is already under version control. Skipping initialization of git.
PS C:\Users\shado\OneDrive\Documentos\GitHub\Practicas_web_Diego_Efrain_Hernandez_Hernandez\Practica 09> cd frontend
PS C:\Users\shado\OneDrive\Documentos\GitHub\Practicas_web_Diego_Efrain_Hernandez_Hernandez\Practica 09\frontend> npm install bootstrap

added 2 packages, and audited 491 packages in 2s

111 packages are looking for funding
  run `npm fund` for details

7 moderate severity vulnerabilities

To address issues that do not require attention, run:
  npm audit fix
```

```
esbuild frontend X TS app.config.ts TS main.ts TS main.server.ts app.html TS app.ts TS tabla.ts tabla.html + v [] ...

PS C:\Users\shado\OneDrive\Documentos\GitHub\Practicas_web_Diego_Efrain_Hernandez_Hernandez\Practica 09\frontend> npm install bootstrap

added 2 packages, and audited 491 packages in 2s

111 packages are looking for funding
  run `npm fund` for details

7 moderate severity vulnerabilities

To address issues that do not require attention, run:
  npm audit fix

To address all issues possible (including breaking changes), run:
  npm audit fix --force

Some issues need review, and may require choosing
a different dependency.

Run `npm audit` for details.
PS C:\Users\shado\OneDrive\Documentos\GitHub\Practicas_web_Diego_Efrain_Hernandez_Hernandez\Practica 09\frontend>
PS C:\Users\shado\OneDrive\Documentos\GitHub\Practicas_web_Diego_Efrain_Hernandez_Hernandez\Practica 09\frontend> ng generate service s
ervices/alumnos

Would you like to share pseudonymous usage data about this project with the Angular Team
at Google under Google's Privacy Policy at https://policies.google.com/privacy. For more
details and how to change this setting, see https://angular.dev/cli/analytics.

No
Global setting: enabled
Local setting: disabled
Effective status: disabled
CREATE src/app/services/alumnos.spec.ts (342 bytes)
CREATE src/app/services/alumnos.ts (120 bytes)
PS C:\Users\shado\OneDrive\Documentos\GitHub\Practicas_web_Diego_Efrain_Hernandez_Hernandez\Practica 09\frontend> ng generate component
formulario
CREATE src/app/formulario/formulario.spec.ts (582 bytes)
CREATE src/app/formulario/formulario.ts (213 bytes)
CREATE src/app/formulario/formulario.css (0 bytes)
CREATE src/app/formulario/formulario.html (26 bytes)
PS C:\Users\shado\OneDrive\Documentos\GitHub\Practicas_web_Diego_Efrain_Hernandez_Hernandez\Practica 09\frontend> ng generate component
tabla
CREATE src/app/tabla/tabla.spec.ts (547 bytes)
CREATE src/app/tabla/tabla.ts (193 bytes)
CREATE src/app/tabla/tabla.css (0 bytes)
CREATE src/app/tabla/tabla.html (21 bytes)
```


VISUAL FRONTEND

```
esbuild frontend  app.html TS app.ts  TS tabla.ts  tabla.html
frontend > src > app > TS app.ts > AppComponent
1  import { Component } from '@angular/core';
2  import { CommonModule } from '@angular/common';
3  import { FormularioComponent } from '../formulario/formulario';
4  import { TablaComponent } from '../tabla/tabla';
5
6  @Component({
7    selector: 'app-root',
8    standalone: true,
9    imports: [CommonModule, FormularioComponent, TablaComponent],
10   templateUrl: './app.html'
11 })
12 export class AppComponent {
13   title = 'P9-frontend';
14 }
```

```
esbuild frontend  app.html  TS app.ts  TS
frontend > src > app > app.html > app-tabla
Go to component
1  <app-formulario></app-formulario>
2  <app-tabla></app-tabla>
```

VISUAL FRONTEND

frontend > src > app > tabla > TS tabla.ts > TablaComponent

```
1 import { Component, OnInit } from '@angular/core';
2 import { CommonModule } from '@angular/common';
3 import { AlumnosService } from '../services/alumnos';
4
5 @Component({
6   selector: 'app-tabla',
7   standalone: true,
8   imports: [CommonModule],
9   templateUrl: './tabla.html'
10 })
11 export class TablaComponent implements OnInit {
12   alumnos: any[] = [];
13
14   constructor(private alumnosService: AlumnosService) {}
15
16   ngOnInit() {
17     this.cargarAlumnos();
18   }
19
20   cargarAlumnos() {
21     this.alumnosService.obtenerAlumnos().subscribe(data => {
22       this.alumnos = data;
23     });
24   }
25 }
```

frontend > src > app > tabla > tabla.html > table.table.table-dark.table-striped

Go to component

```
1 <table class="table table-dark table-striped">
2   <thead>
3     <tr>
4       <th>Nombre</th>
5       <th>Edad</th>
6       <th>Carrera</th>
7     </tr>
8   </thead>
9   <tbody>
10    <tr *ngFor="let alumno of alumnos">
11      <td>{{ alumno.nombre }}</td>
12      <td>{{ alumno.edad }}</td>
13      <td>{{ alumno.carrera }}</td>
14    </tr>
15  </tbody>
16 </table>
```

VISUAL FRONTEND

frontend > src > app > formulario > TS formulario.ts > FormularioComponent

```
1 import { Component } from '@angular/core';
2 import { FormsModule, NgForm } from '@angular/forms';
3 import { CommonModule } from '@angular/common';
4 import { AlumnosService } from '../services/alumnos';
5
6 @Component({
7   selector: 'app-formulario',
8   standalone: true,
9   imports: [CommonModule, FormsModule],
10   templateUrl: './formulario.html'
11 })
12 export class FormularioComponent {
13
14   constructor(private alumnosService: AlumnosService) {}
15
16   guardarAlumno(form: NgForm) {
17     if (form.invalid) return;
18
19     const nuevoAlumno = {
20       nombre: form.value.nombre,
21       edad: form.value.edad,
22       carrera: form.value.carrera
23     };
24
25     this.alumnosService.agregarAlumno(nuevoAlumno).subscribe({
26       next: () => {
27         alert('Alumno insertado correctamente');
28         form.reset();
29       },
30       error: (err) => {
31         console.error('Error al insertar:', err);
32       }
33     });
34   }
35 }
```

esbuild frontend

formulario.html

alumnos.ts

frontend > src > app > formulario > formulario.html > form

Go to component

```
1 <form #formAlumno="ngForm" (ngSubmit)="guardarAlumno(formAlumno)">
2   <div class="mb-3">
3     <label for="nombre">Nombre:</label>
4     <input type="text" class="form-control" ngModel id="nombre" name="nombre" required>
5   </div>
6   <div class="mb-3">
7     <label for="edad">Edad:</label>
8     <input type="number" class="form-control" ngModel id="edad" name="edad" required>
9   </div>
10  <div class="mb-3">
11    <label for="carrera">Carrera:</label>
12    <input type="text" class="form-control" ngModel id="carrera" name="carrera" required>
13  </div>
14  <button type="submit" class="btn btn-primary">Guardar</button>
15 </form>
```


VISUAL FRONTEND

```
frontend > src > app > services > TS alumnos.ts > AlumnosService
1  import { Injectable } from '@angular/core';
2  import { HttpClient } from '@angular/common/http';
3  import { Observable } from 'rxjs';
4
5  @Injectable({
6    providedIn: 'root'
7  })
8  export class AlumnosService {
9    private API = 'http://127.0.0.1:5000/alumnos';
10
11    constructor(private http: HttpClient) {}
12
13    obtenerAlumnos(): Observable<any> {
14      return this.http.get(this.API);
15    }
16
17    agregarAlumno(alumno: any): Observable<any> {
18      return this.http.post(this.API, alumno);
19    }
20  }
```

PÁGINA

▼

Área de trabajo

♥

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Practica_09_Inserciones.pdf - Goog

P9: GET Y POST EN ANGULAR/FLA

Frontend

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📄

localhost:4200

🔍

☆

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👤

...

📁

Ch

Nombre:

Edad:

Carrera:

Guardar

Nombre	Edad	Carrera
Efra GG	21	TIID
Pacap	19	TIID
JLuis Smash	22	TIID
diego	14	Electromecánica
xochitl	15	Maistra

PÁGINA

← ↻ ⓘ localhost:4200 🔍 ☆ ⋮ 🗨 Chat

Nombre:

si

Edad:

45

Carrera:

Lic CPAP

Guardar

localhost:4200 dice

Alumno insertado correctamente

Aceptar

Nombre	Edad	Carrera
Efra GG	21	TIID
Pacap	19	TIID
JLuis Smash	22	TIID
diego	14	Electromecánica
xochitl	15	Maistra

PÁGINA

← ↻ ⓘ localhost:4200 🔍 ☆ ⚙️ 🌐 ⋮ 🗨 Chat

Nombre:

Edad:

Carrera:

Guardar

Nombre	Edad	Carrera
Efra GG	21	TIID
Pacap	19	TIID
JLuis Smash	22	TIID
diego	14	Electromecánica
xochitl	15	Maistra
si	45	Lic CPAP

MONGO

MongoDB Compass - upp_db/escuelaDB.alumnos

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CONNECTIONS (1)

Search connections

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 - eventos
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 - escuelaDB
 - alumnos
 - local

cartelera_cultural alumnos +

upp_db > escuelaDB > alumnos

Documents 3 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA UPDATE DELETE EXPORT DATA EXPORT CODE

25 1 - 6 of 6

1

```
{ "_id": ObjectId("699889b978218561c6d2cd5b"), "nombre": "Pacap", "edad": 19, "carrera": "TIID" }
```

```
{ "_id": ObjectId("699889cd78218561c6d2cd5d"), "nombre": "JLuis Smash", "edad": 22, "carrera": "TIID" }
```

```
{ "_id": ObjectId("699893edbd1253f640b837fa"), "nombre": "diego", "edad": 14, "carrera": "Electromecánica" }
```

```
{ "_id": ObjectId("6998948bbd1253f640b837fb"), "nombre": "xochitl", "edad": 15, "carrera": "Maestra" }
```

```
{ "_id": ObjectId("6998981a5fcb56c5f58eb84d"), "nombre": "si", "edad": 45, "carrera": "Lic CPAP" }
```